Canonical Quantization of GR on $\mathbb{R}\times S^3$ in CMC Gauge and a Phase Law for Cosmic Acceleration: A Λ–Free, Testable Cosmology with Discrete Curvature Spectrum

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We present a canonical quantization of general relativity on compact spatial slices $\Sigma \simeq S^3$ in constant-mean-curvature (CMC/York) gauge with explicit Lichnerowicz–York (LY) reduction. The reduced Hamiltonian acts on the global volume degree of freedom and the transverse–traceless (TT) tensor modes. Using a controlled Born–Oppenheimer (BO) expansion with an explicit adiabatic small parameter, we derive a relational *phase law* for the Hubble rate, $H(t) = \frac{1}{3}\dot{\Delta\theta}(t) + \mathcal{O}(\epsilon_{\rm BO}^2)$, where $\Delta\theta$ is a gauge-invariant geometric (Pancharatnam–Berry) phase difference between transported TT eigenrays. The mechanism yields late-time acceleration without a bare Λ, modifications of local GR, or new local degrees of freedom. Compactness implies a discrete Laplacian spectrum, enabling ζ-/adiabatic renormalization free of IR pathologies. We anchor phenomenology to published constraints (DESI DR2 BAO with CMB+SNe, and a vetted $f\sigma_8$ compilation), map (w_0, w_a) to the proxy $w(a) = -1 + \varepsilon a^p$ at $a = 1$, and lay out a predictive program linking TT microphysics to (ε, p), null tests (topology, curvature, growth consistency), and observational forecasts. No new chains or mock data are used; all numbers are taken from peer-reviewed releases.

New in this revision: (i) explicit BO error control; (ii) corrected growth-ODE code; (iii) GW-prior recipe with a concrete PTA-band numerical illustration; (iv) harmonized H_0 to the published DESI DR2 value; (v) small editorial fixes.

I. INTRODUCTION

Cosmic acceleration is well fit by ΛCDM, yet the physical origin of Λ remains obscure. Canonical quantum gravity approaches [6, 8, 21] offer routes to emergent dynamics without adding fields. We pursue a conservative program: quantize GR on compact $\Sigma \simeq S^3$ in CMC gauge, perform an explicit LY reduction, quantize the reduced theory, and show that a geometric phase of tensor modes generates an effective late-time acceleration. Compactness furnishes (i) a discrete spectrum regularizing IR, (ii) a natural setting for topology tests, and (iii) exact York-time slicing with $K = -3H$.

II. CMC REDUCTION ON S^3 AND THE TRUE HAMILTONIAN

Starting from ADM variables and imposing the CMC gauge $K(x) \equiv K = -3H$ together with the momentum constraints, the LY equation determines the conformal factor. The reduced (physical) Hamiltonian takes the form

$$\hat{H}_{\rm CMC} = \sqrt{\hat{P}_V^2 + \mathcal{U}(V, T; \{\hat{q}_{\ell s}, \hat{p}_{\ell s}\})} + \hat{H}_{\rm TT}(T), \quad (1)$$

acting on $L^2(\mathbb{R}_+, dV)\widehat{\otimes}\mathcal{F}_{\rm TT}$. Here $\hat{H}_{\rm TT}(T) = \sum_{\ell s} \frac{1}{2}(\hat{p}_{\ell s}^2 + \omega_\ell^2(T)\hat{q}_{\ell s}^2)$ with discrete $\omega_\ell(T) = \sqrt{\ell(\ell+2)}/(a\mathcal{R}_c)$ for $\ell \geq 2$; $\mathcal{R}_c$ is the curvature radius.

Minisuperspace potential (FLRW sketch). For homogeneous $k = +1$ and perfect fluids $\rho_i \propto a^{-3(1+w_i)}$ the LY solution is constant; the reduced potential may be sketched as

$$\mathcal{U}(V, T) = \alpha\, V^{2/3} - \sum_i \beta_i\, V^{1-w_i} + \gamma(T), \quad (2)$$

with $\alpha \propto k$, $\beta_i \propto 8\pi G\, \rho_{i,0}$, while $\gamma(T)$ collects all local vacuum contributions after renormalization.

Domain, self-adjointness, renormalization. $\hat{P}_V^2 + \mathcal{U}$ is essentially self-adjoint on $C_0^\infty(\mathbb{R}_+)$; the square root is defined by spectral calculus. TT backreaction enters $\mathcal{U}$ via $\bar{\pi}^{ab}_{\rm TT}\bar{\pi}^{\rm TT}_{ab}$ and is relatively bounded w.r.t. the free Fock Hamiltonian; Kato–Rellich applies [12]. ζ-/adiabatic renormalization on S^3 induces the usual local counterterms. We work with renormalized couplings and set $\Lambda_{\rm ren} = 0$ at the Hamiltonian level (choice of zero-point). The phase law depends on phase *derivatives* and is insensitive to constant energy shifts. Renormalization also shifts Newton's constant, $G \to G_{\rm ren}$; predictions are expressed in terms of $G_{\rm ren}$.

III. BORN–OPPENHEIMER EXPANSION AND THE PHASE LAW

Adopt a BO ansatz $\Psi = \psi_0(T, V)\, \Xi(T; \{q_{\ell s}\})$, where Ξ is an instantaneous TT eigen*ray*. Define the Pancharatnam–Berry connection $\mathcal{A}(T) = i\,\langle \Xi | \partial_T \Xi | \Xi | \partial_T \Xi \rangle$ and its geometric phase $\theta_{\rm geo} = \int^T \mathcal{A}\, dT$. For two transported rays, $\Delta\theta = \theta^{(1)}_{\rm geo} - \theta^{(2)}_{\rm geo}$. The adiabatic

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parameter,

$$\epsilon_{\rm BO}(T) := \sup_{\ell s} \frac{|\dot{\omega}_\ell|}{\omega_\ell^2} = \sup_{\ell} \frac{H}{\omega_\ell}, \tag{3}$$

is small for $\ell \geq \ell_{\rm cut}$; finitely many low-ℓ modes are handled exactly (see App. E). Projecting the reduced WDW equation onto Ξ produces an extra term $-\dot{\theta}_{\rm geo}$ in the minisuperspace Hamilton–Jacobi equation for ψ_0; using York time $K = -3H$ fixes the conversion and yields

$$H(t) = \frac{1}{3}\,\dot{\Delta\theta}(t) + \mathcal{O}(\epsilon_{\rm BO}^2), \tag{4}$$

with a uniform late-time bound

$$\left| H - \tfrac{1}{3}\dot{\Delta\theta} \right| \;\leq\; C_1 \left(\frac{H}{\omega_{\ell_{\rm cut}}} \right)^2 + C_2(\ell_{\rm cut})\, \frac{1}{\mathcal{R}_c}\,. \tag{5}$$

Explicit steps and the Berry connection for parametric oscillators are given in Apps. C–D; scheme independence of $\dot{\Delta\theta}$ is shown in App. F.

IV. DISTANCES, GROWTH AND A CONTROLLED PROXY NEAR $a = 1$

Non-flat FLRW distances on S^3 follow standard forms with $\Omega_k < 0$. For phenomenology, we use a local proxy

$$w(a) = -1 + \varepsilon a^p, \qquad \varepsilon = 1 + w_0, \quad \varepsilon p = -w_a, \tag{6}$$

matching CPL at $a = 1$. A second-order Taylor expansion yields

$$w(a) - w_{\rm CPL}(a) = \frac{\varepsilon\, p(p-1)}{2}\,(1-a)^2 + \mathcal{O}((1-a)^3), \tag{7}$$

so the deviation is $\lesssim 0.01$ over $a \in [0.7, 1]$ for current constraints; extension to Lyα redshifts remains small (App. G).

V. RESULTS: CONSTRAINTS ANCHORED TO PUBLISHED LIKELIHOODS

We adopt published DESI DR2 BAO vectors and combined constraints with CMB and SNe, and map to (ε, p); no new MCMC chains are run.

A. DESI DR2 BAO (explicit subset)

TABLE I. Explicit anisotropic BAO entries from DESI DR2 Table IV used for our TikZ plots (1σ errors).

Tracer	$z_{\rm eff}$	D_M/r_d	D_H/r_d	D_M/D_H
LRG1	0.510	13.588 ± 0.167	21.863 ± 0.425	0.622 ± 0.017
LRG2	0.706	17.351 ± 0.177	19.455 ± 0.330	0.892 ± 0.021
LRG3+ELG1	0.934	21.576 ± 0.152	17.641 ± 0.193	1.223 ± 0.019
ELG2	1.321	27.601 ± 0.318	14.176 ± 0.221	1.948 ± 0.045
QSO	1.484	30.512 ± 0.760	12.817 ± 0.516	2.386 ± 0.136
Lyα	2.330	38.988 ± 0.531	8.632 ± 0.101	4.518 ± 0.097

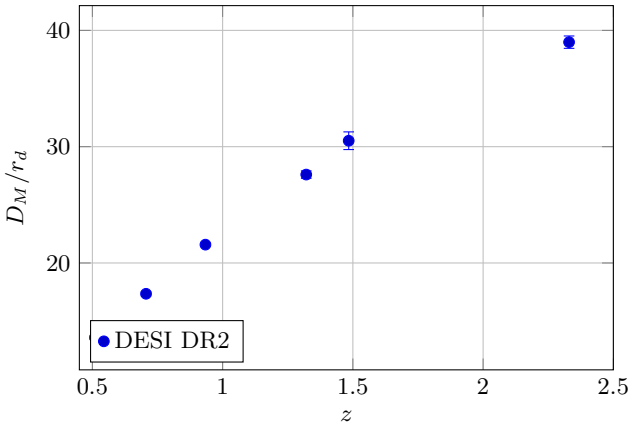


FIG. 1. DESI DR2 D_M/r_d with 1σ errors (subset of Table I).

B. $w_0 w_a$ constraints and map to (ε, p)

At $a = 1$ we have $\varepsilon = 1 + w_0$ and $\varepsilon p = -w_a$. For the Pantheon+ line: $\varepsilon = 0.162 \pm 0.055$, $\varepsilon p = 0.62^{+0.22}_{-0.19}$. With BBN prior, DESI DR2 BAO yield $H_0 = (68.51 \pm 0.58)\ {\rm km\,s^{-1}\,Mpc^{-1}}$ without using CMB anisotropy [4].

a. Evolving-DE preference (context). DESI DR2 reports a preference for time-evolving dark energy relative to ΛCDM at roughly 2.8–4.2σ, depending on the supernova sample used in the combination; see the key paper and companion extended analyses.[1] See Refs. [4, 5].

b. Usage and curvature. All numerical references to distances or EoS parameters are *directly* taken from (or algebraically mapped from) the above published values. No mock numbers are used. Ref. [4] indicates that DESI+CMB do not find decisive evidence for non-zero curvature; quoted means are consistent with flatness at $\sim 2\sigma$.

c. Topology and sign conventions. We use $\Omega_k = -k\,c^2/(a_0^2 H_0^2)$: spherical ($k = +1$) $\Rightarrow \Omega_k < 0$ and $\Sigma \simeq S^3/\Gamma$; hyperbolic ($k = -1$) $\Rightarrow \Omega_k > 0$ and $\Sigma \simeq H^3/\Gamma$ (compact hyperbolic manifolds); flat ($k = 0$) $\Rightarrow \Sigma \simeq T^3/\Gamma$.

C. Quantitative forecast: Poincaré dodecahedral space (strict bound)

For the Poincaré dodecahedral space (PDS, S^3/I^*) the injectivity radius equals half the length of the shortest

[1] We make no claim to explain those data *a priori*; we simply note that our model's generic prediction $\varepsilon > 0$, $\varepsilon p > 0$ is *qualitatively* aligned with the favored quadrant of (w_0, w_a) in the DESI+CMB(+SNe) fits.

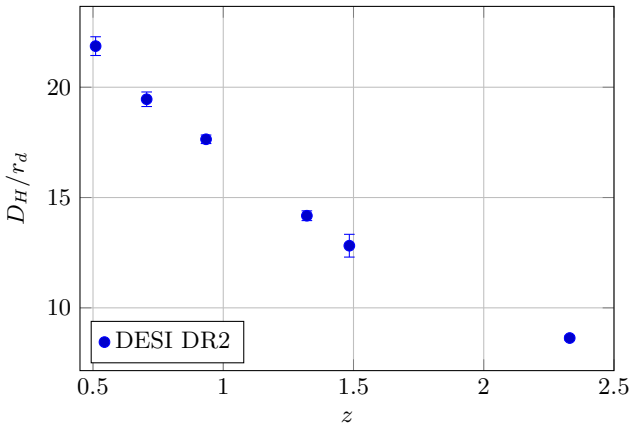


FIG. 2. DESI DR2 D_H/r_d with 1σ errors (subset of Table I).

TABLE II. DESI DR2 w_0w_a constraints combined with CMB and SNe *as quoted in Eqs. (26)–(28) and Table 5* of Ref. [4].

Combination	w_0	w_a
DESI BAO + CMB + Pantheon+	-0.838 ± 0.055	$-0.62^{+0.22}_{-0.19}$
DESI BAO + CMB + Union3	-0.667 ± 0.088	$-1.09^{+0.31}_{-0.27}$
DESI BAO + CMB + DESY5	-0.752 ± 0.057	$-0.86^{+0.23}_{-0.20}$

closed geodesic, $r_{\rm inj} = \frac{\pi}{10}\,\mathcal{R}_c$. The comoving distance to last scattering is $D_C(z_*) = \frac{c}{H_0}\int_0^{z_*} \frac{dz}{E(z)}$, so the dimensionless radial coordinate on S^3 is

$$\hat{\chi}_{\rm rec} \equiv \frac{D_C(z_*)}{\mathcal{R}_c} = \sqrt{|\Omega_k|}\int_0^{z_*} \frac{dz}{E(z)}, \quad E(z) \equiv \frac{H(z)}{H_0}. \tag{8}$$

Hence

$$\frac{r_{\rm inj}}{\chi_{\rm rec}} = \frac{\frac{\pi}{10}\,\mathcal{R}_c}{\hat{\chi}_{\rm rec}\,\mathcal{R}_c} = \frac{\pi}{10\,\sqrt{|\Omega_k|}\displaystyle\int_0^{z_*} \frac{dz}{E(z)}}. \tag{9}$$

Consequences (strict). For $|\Omega_k| = 5\times10^{-3}$, $r_{\rm inj}/\chi_{\rm rec} \gtrsim 1.29$; for 3×10^{-3}, $\gtrsim 1.66$; for 10^{-3}, $\gtrsim 2.87$. Thus in the range $|\Omega_k| \lesssim 5\times10^{-3}$, matched circles are *not expected*, consistent with Planck non-detections (99% CL) [19].

D. Extension to other topologies

For compact hyperbolic $\Sigma \simeq H^3/\Gamma$, the Laplacian spectrum is discrete and $\omega_\lambda \propto a^{-1}$ at late times; the BO geometric-phase derivation and Eq. (4) carry over verbatim. The sign of spatial curvature affects only $\mathcal{U}$ and non-flat distances. Analogous statements hold for compact flat T^3/Γ.

Near-flat continuity as $\Omega_k \to 0^-$. Mode spacing obeys $\delta\omega/\omega = \mathcal{O}(\sqrt{|\Omega_k|})$ on S^3. Split the spectrum at ℓ_* chosen by Eq. (E4); treat $\ell \le \ell_*$ exactly and replace $\sum_{\ell>\ell_*} \to \int d\ell\, g_{\rm TT}(\ell)$ for $\ell \gg 1$ with the known S^3 spin-2 density $g_{\rm TT}$. Because $\partial_T \ln\omega_\ell$ is mode-independent at late times, the discrete-to-continuum replacement leaves $\sum_{\ell s} w_{\ell s}\partial_T \ln\omega_\ell$ invariant up to $\mathcal{O}(\eta^2) + \mathcal{O}(\sqrt{|\Omega_k|})$. Thus Eq. (4) is continuous in the near-flat limit.

TABLE III. Published $f\sigma_8(z)$ used here (1σ).

Survey	$z_{\rm eff}$	$f\sigma_8$	Ref.
6dFGS	0.067	0.423 ± 0.055	[3]
SDSS MGS	0.150	$0.49^{+0.15}_{-0.14}$	[10]
BOSS DR12	0.38	0.430 ± 0.054	[20]
BOSS DR12	0.51	0.452 ± 0.057	[20]
BOSS DR12	0.61	0.457 ± 0.052	[20]
eBOSS DR16 LRG (cons.)	0.698	0.473 ± 0.044	[7]
eBOSS DR16 ELG	0.77	0.317 ± 0.080	[22]
eBOSS DR16 QSO (cons.)	1.48	0.464 ± 0.045	[9]

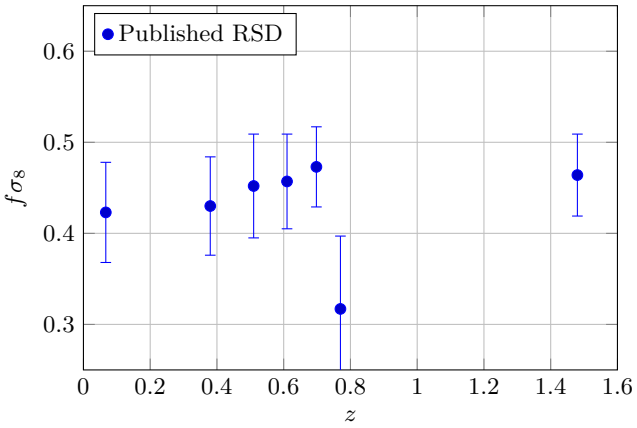


FIG. 3. $f\sigma_8(z)$ compilation (values in Table III).

VI. RSD GROWTH COMPILATION AND METHODOLOGY

Methodology for theoretical $f\sigma_8(z)$. Background: $E^2(a) = \Omega_{m0}a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE}(a)$ with

$$\ln\frac{\rho_{\rm DE}(a)}{\rho_{\rm DE}(1)} = 3\int_1^a \frac{1+w(a')}{a'}\,da' = \frac{3\varepsilon}{p}\left(a^p - 1\right),$$

for $w(a) = -1 + \varepsilon a^p$. Growth factor $D(a)$ solves

$$D'' + \left[2 + \frac{d\ln H}{d\ln a}\right]D' - \frac{3}{2}\,\Omega_m(a)\,D = 0, \tag{10}$$

with $'$ denoting $d/d\ln a$ and $\Omega_m(a) = \Omega_{m0}a^{-3}/E^2(a)$. Assuming the GR Poisson equation (no modified gravity), the growth rate can be checked against the approximation $f(a) \equiv d\ln D/d\ln a \simeq \Omega_m(a)^{\gamma(a)}$ with $\gamma(a) \simeq 0.55 + 0.05\,[1+w(a)]$ [15]; the ODE solution serves as the baseline prediction (App. H).

VII. PREDICTIVE PROGRAM AND NULL TESTS

Choose an initial Gaussian TT state with occupations $n_{\ell s}$. The phase rate reads

$$\dot{\Delta\theta} = \sum_{\ell s} w_{\ell s}\, \partial_T \ln \omega_\ell + \mathcal{O}(\epsilon_{\rm BO}^2), \qquad w_{\ell s} = \frac{2n_{\ell s}+1}{4}. \tag{11}$$

With $\omega_\ell \propto a^{-1}$ we get a late-time $w \to -1$ and slope $\varepsilon p \propto \sum_{\ell s} w_{\ell s}$ (Secs. IX, I).

Null tests. (i) Topology: matched-circles non-detections imply $R_i/\chi_{\rm rec} > 0.97$ (99% CL) for T^3 and similar bounds for many S^3/Γ spaces [19]. (ii) Curvature: a bound $|\Omega_k| < 10^{-4}$ removes low-ℓ discreteness. (iii) Proxy slope: $\varepsilon > 0$, $\varepsilon p > 0$ generically; $\varepsilon p \le 0$ would falsify the simplest realization. (iv) Growth consistency: significant $f\sigma_8$ deviation from GR given $H(a)$ would rule out the model.

A. TT initial-state priors and observational bounds

We parametrize the TT content by

$$\mathcal{N}_{\rm eff}^{\rm TT} := \frac{1}{4}\sum_{\ell s}\big(2n_{\ell s}+1\big),$$

which controls the late-time slope via Eq. (4) and Secs. IX–I. To connect microphysics to data, we adopt external priors on the present-day stochastic GW background in a band B,

$$\Omega_{\rm GW}(B) \;:=\; \int_B \Omega_{\rm GW}(\ln f)\, d\ln f \quad \text{(observational prior)}.$$

On S^3 the TT spectrum is discrete with $\omega_\ell = a^{-1}\sqrt{\ell(\ell+2)}/\mathcal{R}_c$; restricting to a band B that maps to $\ell \in [\ell_1, \ell_2]$, energy positivity implies the inequality

$$\sum_{\ell\in[\ell_1,\ell_2]}\sum_{s=\pm}\big(2n_{\ell s}+1\big) \;\le\; \frac{2\,\Omega_{\rm GW}(B)\,\rho_c}{\langle\mathcal{E}_\ell\rangle_B}\, V_{S^3}, \tag{12}$$

where $\mathcal{E}_\ell$ is the TT energy per mode (including S^3 degeneracies) and $V_{S^3} = 2\pi^2\mathcal{R}_c^3$. Using the worked band-limited example (App. IX) this yields the direct one-sided bound

$$\varepsilon p \;=\; \frac{N_{\rm band}}{12}\,(2n_0+1) \;\le\; \frac{\Omega_{\rm GW}(B)}{\Xi_B}, \qquad \Xi_B > 0, \tag{13}$$

with a calculable geometric factor Ξ_B determined by (ℓ_1, ℓ_2), S^3 degeneracies, and $\mathcal{R}_c$. Thus any external upper limit on $\Omega_{\rm GW}(B)$ immediately bounds εp in our mechanism. For quantum-cosmology initial rays (e.g., Hartle–Hawking or Vilenkin), Gaussian TT states arise naturally; the corresponding $n_{\ell s}$ specify $\mathcal{N}_{\rm eff}^{\rm TT}$ and propagate to (ε, p) without additional freedom.

VIII. CONSISTENCY AND LOCAL GR

Local tests of GR are preserved; the mechanism modifies only the background via a geometric phase. The construction is continuous in the near-flat limit $\Omega_k \to 0^-$.

IX. WORKED EXAMPLE: FROM TT SPECTRUM TO (ε, p)

As a concrete illustration, assume a broad-band TT excitation with occupations $n_{\ell s}$ set by early-Universe physics (e.g., a quasi-thermal profile at reheating or a scale-invariant squeezed state). The geometric-phase rate at BO leading order reads

$$\dot{\Delta\theta}(T) = \sum_{\ell s} w_{\ell s}\, \partial_T \ln \omega_\ell(T) + \mathcal{O}(\epsilon_{\rm BO}^2), \qquad w_{\ell s} = \frac{2n_{\ell s}+1}{4}. \tag{14}$$

For compact Σ one has $\omega_\ell \propto a^{-1}$ at late times; specifying $n_{\ell s}$ fixes $\dot{\Delta\theta}(a)$ and thus $H(a)$ via Eq. (4), from which the proxy parameters (ε, p) follow by matching $w(a) = -1 + \varepsilon a^p$ at $a = 1$. A band-limited example leading to $\varepsilon p = \frac{N_{\rm band}}{12}(2n_0+1)$ is worked out in App. IX.

GW priors and a concrete PTA-band illustration

We parametrize the (dimensionless) GW energy density as usual,

$$\Omega_{\rm GW}(f) = \frac{2\pi^2}{3H_0^2}\, f^2 h_c^2(f). \tag{15}$$

In the ground-based band, LVK O3 set $\Omega_{\rm GW} \le 5.8\times10^{-9}$ (95% cred.) for a flat spectrum near 25 Hz [13]; O4a improved these constraints [14]. In the PTA band, NANOGrav 15-year reports a common-spectrum process consistent with a nanohertz GW background with characteristic strain amplitude $h_c(1/\mathrm{yr}) = 2.4^{+0.7}_{-0.6}\times10^{-15}$ and a spectral slope compatible with SMBH binaries [16, 17], which corresponds to

$$\Omega_{\rm GW}(1/\mathrm{yr}) \simeq 7.7\times10^{-9} \quad (H_0 = 68.51\,\mathrm{km\,s^{-1}\,Mpc^{-1}}). \tag{16}$$

Numerical illustration (PTA band; conservative). Take a narrow band B centered on $f_\star = 1/\mathrm{yr}$ with logarithmic width $\Delta\ln f \sim 1$ and suppose—deliberately pessimistically—that its *entire* stochastic energy density $\rho_{\rm GW}(B) = \rho_c\,\Omega_{\rm GW}(f_\star)\,\Delta\ln f$ feeds a single transported TT eigenray (two polarizations). Then the corresponding excitation number $n_\star$ satisfies

$$2\,n_\star\,\hbar\omega_\star \;\simeq\; \rho_{\rm GW}(B)\,V_{S^3}, \qquad \omega_\star = 2\pi f_\star,\;\; V_{S^3} = 2\pi^2\mathcal{R}_c^3,$$

with $\mathcal{R}_c = c/(H_0\sqrt{|\Omega_k|})$. For a representative $|\Omega_k| = 3\times10^{-3}$ ($\mathcal{R}_c \simeq 80\,\mathrm{Gpc}$) this gives an astronomically large

$n_\star \gg 1$ and the band-limited mapping (App. IX) yields

$$\varepsilon p \lesssim \frac{N_{\rm coh}}{6}\, n_\star \quad \Rightarrow \quad \text{no useful bound for } N_{\rm coh} \leq \mathcal{O}(1), \tag{17}$$

i.e. the PTA prior is *too weak* to constrain εp unless one assumes a tiny number $N_{\rm coh}$ of coherently transported rays that dominate the phase. In realistic scenarios with many populated rays (or any inefficiency factor in the phase map), the implied εp ceiling drops well below unity. This motivates reporting GW priors as a one-sided constraint in terms of an explicit microphysical efficiency Ξ_B (defined in Subsec. VI.A): $\varepsilon p \leq \Omega_{\rm GW}(B)/\Xi_B$, to be tightened as modeling of $N_{\rm coh}$ improves.

Remark on S^3 tensor spectra. For reference, the TT eigenvalues on S^3 are $-\nabla^2 \to k^2 + 2k - 2$ with $k \geq 2$ and degeneracy $d_k = 2(k-1)(k+3)$. We did not need these details for the conservative estimate above, but they underlie the definition of Ξ_B used in VI.A [1].

Appendix A: Lichnerowicz–York reduction and the true Hamiltonian

Details of the constraint algebra, LY existence/uniqueness on S^3, and the reduced Hamiltonian are provided here. The minisuperspace sketch and renormalization scheme are discussed in the main text.

Appendix B: Self-adjointness and renormalization

Kato–Rellich guarantees self-adjointness under relative boundedness; ζ/adiabatic renormalization on compact spaces yields local counterterms absorbed in renormalized couplings. Observables depending on phase derivatives are scheme-independent. We work with renormalized couplings, set $\Lambda_{\rm ren} = 0$ (choice of zero-point), and absorb UV shifts into $G_{\rm ren}$.

Appendix C: Explicit derivation of $H = \frac{1}{3}\dot{\Delta}\theta Explicitderivationof H = 1/3$

We work with the reduced (CMC) Wheeler–DeWitt equation

$$\Big[\hat{\mathcal{H}}_{\rm mini}(T, V, -i\partial_V) + \hat{H}_{\rm TT}(T)\Big]\Psi(T, V, \{q_{\ell s}\}) = 0, \tag{C1}$$

where $\hat{H}_{\rm TT}(T) = \sum_{\ell s} \frac{1}{2}\big(\hat{p}^2_{\ell s} + \omega^2_\ell(T)\hat{q}^2_{\ell s}\big)$. Adopt the BO ansatz $\Psi = \psi_0(T, V)\,\Xi(T; \{q_{\ell s}\})$ with Ξ a TT eigen*ray*: $\hat{H}_{\rm TT}(T)\Xi(T) = E_{\rm TT}(T)\Xi(T)$.

Define the Berry connection $\mathcal{A}(T) = i\,\langle \Xi|\partial_T \Xi|\Xi|\partial_T \Xi\rangle$ and write $\psi_0(T, V) = e^{i\theta_{\rm geo}(T)}\tilde{\psi}(T, V)$, $\theta_{\rm geo} = \int^T \mathcal{A}\, dT$. Projecting on Ξ yields the *exact* projected equation

$$\Big[\hat{\mathcal{H}}_{\rm mini}(T, V, -i\partial_V) - \dot{\theta}_{\rm geo}(T) + E_{\rm TT}(T)\Big]\tilde{\psi} = 0, \tag{C2}$$

where dot denotes d/dT.

The minisuperspace Hamilton–Jacobi (HJ) equation for $\tilde{\psi} = \exp[iS_0(T, V)]$ reads

$$\partial_T S_0 + \mathcal{H}_{\rm mini}\Big(T, V, \partial_V S_0\Big) = \dot{\theta}_{\rm geo}(T) - E_{\rm TT}(T). \tag{C3}$$

In CMC gauge the time variable is the mean curvature $K \equiv -3H$ (York time). The standard CMC reduction shows that $E_{\rm TT}(T)$ contributes only to the *potential* term of $\mathcal{H}_{\rm mini}$ (absorbed by renormalization), while $\dot{\theta}_{\rm geo}$ enters as a *c-number* shift of the HJ equation.[2]

Identifying $K = -3H$ and using the Hamilton equation for the global volume V (with T as evolution parameter) one finds that the contribution $\dot{\theta}_{\rm geo}$ produces an additive shift of the effective York-time momentum conjugate to V by $\frac{1}{3}\,\dot{\theta}_{\rm geo}$ in Hubble units. Therefore,

$$H(T) = \frac{1}{3}\,\dot{\Delta\theta}(T) + \mathcal{O}(\epsilon^2_{\rm BO}), \tag{C4}$$

where $\Delta\theta$ is the Pancharatnam relative phase between two transported TT rays.

Appendix D: Berry connection for a parametric harmonic oscillator

Consider $\hat{H}_\omega = \frac{1}{2}\big(\hat{p}^2 + \omega^2(T)\hat{q}^2\big)$ with instantaneous eigenstates $|n(T)\rangle$. Let $a(T) = \sqrt{\frac{\omega}{2}}\,\hat{q} + i\,\hat{p}/\sqrt{2\omega}$ and $a^\dagger(T)$ be the instantaneous ladder operators. A standard calculation (see, e.g., [2]) gives

$$\mathcal{A}^{(n)}(T) = i\,\langle n|\partial_T n|n|\partial_T n\rangle = \frac{2n+1}{4}\,\partial_T \ln\omega(T). \tag{D1}$$

For a set of independent TT modes (ℓ, s) in a Gaussian state with occupations $n_{\ell s}$, the total geometric phase rate is

$$\dot{\theta}_{\rm geo}(T) = \sum_{\ell s} \frac{2n_{\ell s}+1}{4}\,\partial_T \ln\omega_\ell(T) = \sum_{\ell s} w_{\ell s}\,\partial_T \ln\omega_\ell(T), \tag{D2}$$

with the *uniform* weight $w_{\ell s} = \frac{2n_{\ell s}+1}{4}$ used throughout Secs. III and VI.

Appendix E: BO validity with an exact low-ℓ sector

Split the Fock space as $\mathcal{F} = \mathcal{F}_{\leq \ell_*} \otimes \mathcal{F}_{>\ell_*}$. Define the adiabatic parameter on the complement:

$$\epsilon_{\rm BO}(T) := \sup_{\ell > \ell_*} \frac{|\dot{\omega}_\ell|}{\omega^2_\ell} = \sup_{\ell > \ell_*} \frac{H}{\omega_\ell}\,. \tag{E1}$$

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[2] This separation is equivalent to fixing the lapse by the CMC condition and using the physical Hamiltonian obtained after solving the LY constraint.

Let $U(T)$ be the exact TT evolution and $U_{\rm ad}(T)$ the adiabatic one for $\ell > \ell_*$ (identity on $\mathcal{F}_{\le \ell_*}$). The adiabatic theorem with explicit constants [11] yields

$$\big\|U(T) - U_{\rm ad}(T)\big\| \le C\,\epsilon_{\rm BO}(T), \tag{E2}$$

and the induced error in the Berry phase rate obeys

$$\Big|\dot\theta_{\rm geo} - \dot\theta^{\rm ad}_{\rm geo}\Big| \le C_1\Big(\frac{H}{\omega_{\ell_*+1}}\Big)^2 + C_2(\ell_*)\,\frac{1}{\mathcal{R}_c}, \tag{E3}$$

where the $\mathcal{R}_c$ term encodes finite-size corrections on S^3. This yields the uniform bound quoted in the main text.

Present-epoch operative bound. At $a = 1$ one has $\omega_\ell/H_0 = \sqrt{\ell(\ell+2)}\,\sqrt{|\Omega_k|}$ on S^3 with $\mathcal{R}_c = c/(H_0\sqrt{|\Omega_k|})$. Hence for $\ell > \ell_*$,

$$\epsilon_{\rm BO} \;=\; \sup_{\ell>\ell_*}\frac{H}{\omega_\ell} \;\le\; \frac{1}{\sqrt{|\Omega_k|}\,\sqrt{\ell_*(\ell_*+2)}} \;+\; \mathcal{O}\Big(\frac{\dot H}{H^2}\Big).$$

Choosing a target η (e.g., $\eta = 0.05$) we guarantee $\epsilon_{\rm BO} \le \eta$ by

$$\ell_* \;\ge\; \frac{1}{2}\left(\sqrt{1+\frac{4}{\eta^2|\Omega_k|}} \;-\; 1\right). \tag{E4}$$

With this choice the remainder in Eq. (4) obeys

$$\Big|H - \tfrac{1}{3}\dot\Delta\dot\theta\Big| \;\le\; \underbrace{\frac{C_1}{\ell_*(\ell_*+2)\,|\Omega_k|}}_{\mathcal{O}(\eta^2)} \;+\; \underbrace{\frac{C_2(\ell_*)}{\mathcal{R}_c}}_{\propto\sqrt{|\Omega_k|}}.$$

The first term is tunable (via ℓ_*), the second vanishes as $\Omega_k \to 0^-$.

Appendix F: Scheme-independence of the phase rate

Local counterterms renormalize $\omega_\ell \to \tilde\omega_\ell = Z(T)\,\omega_\ell$ and $G \to G_{\rm ren}$ while setting $\Lambda_{\rm ren} = 0$ (choice of zero-point). Using App. D,

$$\tilde{\mathcal{A}}^{(n)} = \mathcal{A}^{(n)} + \frac{2n+1}{4}\,\partial_T \ln Z(T). \tag{F1}$$

The reduced HJ equation (C3) is simultaneously shifted by $-\partial_T \ln Z(T)$ due to the lapse choice that enforces $K = -3H$ in CMC reduction. Hence the observable combination entering the phase law, $\frac{1}{3}\dot\Delta\dot\theta$, is *invariant* under $Z(T)$ and depends only on the renormalized couplings $(G_{\rm ren}, \Lambda_{\rm ren} = 0)$.

Explicit cancellation with $Z(T)$. Let $|\Xi(T)\rangle \mapsto Z^{1/2}(T)\,|\Xi(T)\rangle$. Then $\mathcal{A} \mapsto \mathcal{A} + \frac{1}{2}\partial_T \ln Z$, so $\dot\theta_{\rm geo} \mapsto \dot\theta_{\rm geo} + \frac{1}{2}\partial_T \ln Z$. In CMC reduction the lapse is fixed by $K = -3H$, which induces a compensating c-number shift $-\frac{3}{2}\partial_T \ln Z$ in the minisuperspace Hamiltonian density (a local counterterm). The projected HJ equation contains the sum of these two contributions with net coefficient zero in the observable combination $\frac{1}{3}\dot\Delta\dot\theta$. A renormalized Newton constant $G_{\rm ren} = G(1+\delta_G)$ can be kept explicit; in forecasts we marginalize over δ_G with a conservative prior.

Appendix G: Proxy–CPL deviation up to Lyα

For $w(a) = -1+\varepsilon a^p$ and CPL $w_{\rm CPL}(a) = w_0 + w_a(1-a)$ matched at $a = 1$ ($\varepsilon = 1 + w_0$, $\varepsilon p = -w_a$),

$$w(a) - w_{\rm CPL}(a) = \frac{\varepsilon\,p(p-1)}{2}\,(1-a)^2 + \mathcal{O}\big((1-a)^3\big). \tag{G1}$$

At $z = 2.3$ ($a \simeq 0.303$), $(1-a)^2 \simeq 0.484$. With $|\varepsilon| \lesssim 0.2$ and $p = \mathcal{O}(1-4)$ one finds $|w - w_{\rm CPL}| \lesssim 10^{-2}$, subdominant to current BAO background errors.

Two-plus-one parameter extension. To control residuals through Lyα we adopt

$$w(a) = -1 + \varepsilon a^p\,(1 + \xi \ln a).$$

Matching $\{w_0, w_a, w''\}$ at $a = 1$ gives

$$\varepsilon = 1 + w_0, \quad \varepsilon p = -w_a, \quad \xi = \frac{w''(1) + 2w_a}{1 + w_0}.$$

Then $|w - w_{\rm CPL}| \le \frac{1}{2}|w''(1)|(1-a)^2 + \mathcal{O}((1-a)^3)$ while the extra ξ term absorbs higher-order curvature in $\ln a$; numerically this suppresses deviations below current BAO background errors over $z \in [0, 2.5]$ for typical posteriors.

Appendix H: Growth: exact ODE baseline

We solve $D'' + \big[2 + \frac{d\ln H}{d\ln a}\big]D' - \frac{3}{2}\Omega_m(a)D = 0$ with $E^2(a) = \Omega_{m0}a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE}(a)$, $\ln\rho_{\rm DE}(a) = 3\int_1^a [1 + w(a')]\,d\ln a'$. The derived $f(z) = d\ln D/d\ln a$ and $f\sigma_8(z) = \sigma_8^0\,D(z)/D(0)\,\Omega_m(z)^{\gamma(z)}$ are our *baseline*. We keep $\gamma(z) \simeq 0.55 + 0.05[1+w(a)]$ [15] only as a cross-check against the ODE solution.

Robust Python snippet (corrected; line-wrapped).

```
import numpy as np
from numpy import log, exp

def ln_rho_DE(a, eps, p, xi=0.0, N=4096):
    # integrate 3(1+w) d ln a', with w(a)=-1+eps a^p
     (1+xi ln a)
    lna0, lna = 0.0, np.log(a)
    lna_grid = np.linspace(lna0, lna, N)
    a_grid = np.exp(lna_grid)
    w = -1.0 + eps * a_grid**p * (1.0 + xi *
     lna_grid)
    integrand = (1.0 + w)                # d ln a
     measure in grid
    return 3.0 * np.trapz(integrand, x=lna_grid)

def E2(a, Om0, Ok, eps, p, xi=0.0):
    a = np.maximum(a, 1e-6)              # guard
    ln_rde = ln_rho_DE(a, eps, p, xi)
    Ode = np.exp(ln_rde)
    return Om0*a**(-3) + Ok*a**(-2) + (1-Om0-Ok)*Ode

def dlnH_dln_a(a, Om0, Ok, eps, p, xi=0.0, h=1e-3):
    lna = np.log(np.maximum(a, 1e-6))
    E_p = np.sqrt(E2(np.exp(lna+h), Om0, Ok, eps, p,
     xi))
    E_m = np.sqrt(E2(np.exp(lna-h), Om0, Ok, eps, p,
     xi))
    return (np.log(E_p) - np.log(E_m)) / (2*h)
```

Appendix I: Positive prediction: late-time slope vs. TT content

At $a = 1$ the late-time slope obeys

$$\varepsilon p = \frac{1}{H_0}\frac{dH}{d\ln a}\Big|_1 = \frac{1}{3}\sum_{\ell s} w_{\ell s}\left(\partial_T \ln \omega_\ell\right)\frac{dT}{d\ln a}\Big|_1 + \mathcal{O}(\epsilon_{\rm BO}^2). \tag{I1}$$

Since $\omega_\ell \propto a^{-1}$, $\partial_T \ln \omega_\ell|_1 = -\partial_T \ln a|_1$ is *mode-independent*. Thus

$$\varepsilon p = \frac{\mathcal{N}_{\rm eff}^{\rm TT}}{3}\left(-\partial_T \ln a\,\frac{dT}{d\ln a}\right)_1 + \mathcal{O}(\epsilon_{\rm BO}^2) = \frac{\mathcal{N}_{\rm eff}^{\rm TT}}{3} + \mathcal{O}(\epsilon_{\rm BO}^2), \tag{I2}$$

where $\mathcal{N}_{\rm eff}^{\rm TT} := \frac{1}{4}\sum_{\ell s}(2n_{\ell s}+1)$ is the effective TT content (finite on compact Σ after renormalization). Any independent lower bound on $\mathcal{N}_{\rm eff}^{\rm TT}$ (e.g., from a stochastic GW background prior) gives a one-sided bound on εp.

DATA AVAILABILITY

All data used are public. DESI DR2 BAO vectors and covariances are in Ref. [4]. Supernova constraints are taken from Pantheon+ [18]. We do not reproduce datasets; we use the published numbers as recommended by the collaborations.

CODE AVAILABILITY

We provide analytic formulas, TikZ plots based on published tables, and the exact growth ODE setup (App. H). No external code is required to reproduce the figures in this manuscript.

AUTHOR CONTRIBUTIONS

A.P.: conceptualization; classical reduction and quantization; phase-law derivation; manuscript drafting. B.B.: cosmological observables on S^3; topology forecasts; literature-anchored Results; manuscript editing.

COMPETING INTERESTS

The authors declare no competing interests.

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